HOW TO USE THE BOOK

Scope of Physics
- Awareness on higher learning - courses, institutions and required competitive exams
- Financial assistance possible to help students to climb academic ladder
- Gender initiatives by the Government of India.

Learning Objectives:
- Overview of the unit
- Gives clarity on the goals and objective of the topics



- Additional facts related to the topics covered to facilitate curiosity driven learning

Example problems
- To ensure understanding, problems/illustrations are given at every stage before advancing to next level



- Visual representation of concepts with illustrations
- Videos, animations, and tutorials



ICT
- To harness the digital skills to class room learning and experimenting

Summary
- Recap of salient points of the lesson

Concept Map
- Schematic outline of salient learning of the unit

Evaluation
- Evaluate students' understanding and get them acquainted with the application of physical concepts to numerical and conceptual questions

Books for Reference
- List of relevant books for further reading

Solved examples
- Solutions to exercise problems are accessible here. In addition, a few solved examples are given to facilitate students to apply the concepts learnt.

Competitive Exam corner
- Model Questions - To motivate students aspiring to take up competitive examinations such as NEET, JEE, Physics Olympiad, JIPMER etc

Glossary
- Scientific terms frequently used with their Tamil equivalents

Content Focus
- Covers ray and wave optics extensively, salient concepts in dual nature of radiation and matter, atomic and nuclear physics. Topics in semiconductors and communication are optimised. An exclusive unit on 'Recent developments in physics' highlights that physics is the basic building block for sciences, engineering, technology and medicine. With this, students are motivated to pursue higher education confidently.

Back Wrapper

***Richard Philip Feynman**, (1918–1988) a theoretical physicist who received noble prize in physics in 1965 for his contributions to the development of quantum electrodynamics. He is the first person to discuss the possiblity of manipulation of atoms that seeded nanotechnology. His lectures on various topics in physics are very popular among physicists.*

Illustration of Gravitational waves from two merging black holes.
Actual photograph of a super massive black hole M87*

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